



Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (W), Mumbai: 400058, India

(An Empowered Autonomous College Affiliated to University of Mumbai)

Department of Computer Engineering

Vehicle Rental Management System – Class Diagram

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Aim:

To draw a Class diagram for a Vehicle Rental Management System.

Implementation:

Mentioned below is the chronological order that was followed to design a Class diagram for our vehicle rental management software. Initially, the problem description was re-visited to identify nouns and noun phrases. These were then reviewed to get rid of any ambiguity and classes were identified. Relations between different classes and how they interact with each other were noted to get an idea about the overall structure of the class diagram. Finally, the class diagram was constructed keeping the above-mentioned points in mind.

Problem Description:

A vehicle rental management software needs to be designed for a particular rental agency. This software should be capable of authorising different types of users and allowing them to perform their respective functions. A guest user must be able to browse the catalogue of vehicles through an interface and view the tariffs of the vehicles offered by the agency, such as cars and two-wheelers. A guest user must be able to register himself. A registered customer must be able to log into his account, submit a driving licence for verification and place a booking for a vehicle for a chosen period at a chosen branch. Branch staff would be responsible for handing over a vehicle to the customer and for recording its return, capturing a condition report on each occasion. A customer must be able to manage his bookings in terms of creation, modification, extension and cancellation. A customer must also be able to pay for a booking, receive an invoice and view his rental history. A maintenance technician must be able to record and close maintenance jobs raised against a vehicle. An administrator must be able to manage the fleet, define tariffs and approve refunds. Currently, the agency offers two types of vehicles – Car and TwoWheeler – and accepts three types of payment – Card, Cash and Online.



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Following is a breakdown of various parts that were considered for designing the class diagram -

[1] Noun / Noun Phrases.

Agency, User, Function, Guest User, Registered Customer, Branch Staff, Branch Manager, Maintenance Technician, Administrator, Licence, Booking, Vehicle, Car, Two-wheeler, Branch, Tariff, Payment, Card Payment, Cash Payment, Online Payment, Invoice, Condition Report, Maintenance Job, Interface, Security Deposit, Rental History.

[2] Classes.

User, Customer, BranchStaff, MaintenanceTechnician, Administrator, Licence, BranchManager, Booking, Vehicle, Car, TwoWheeler, Branch, Tariff, Payment, CardPayment, CashPayment, OnlinePayment, Invoice, ConditionReport, MaintenanceJob.

[3] Verb Phrases.

1. Customer registers and logs into the software.
2. Customer submits a Licence for verification.
3. Customer searches Vehicles and views Tariffs.
4. Customer creates, modifies, extends or cancels a Booking.
5. Customer pays for a Booking and receives an Invoice.
6. Customer views rental history.
7. BranchStaff hands over and returns a Vehicle.
8. BranchStaff records a ConditionReport.
9. MaintenanceTechnician records and closes a MaintenanceJob.
10. Administrator manages the fleet, defines Tariffs and approves refunds.



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[4] Relations.

1. Generalization (inheritance): Customer, BranchStaff, MaintenanceTechnician and Administrator are different users of the software and hence are generalized to User.
2. Generalization (inheritance): BranchManager is a restricted Administrator and is therefore generalized to Administrator.
3. Generalization (inheritance): Car and TwoWheeler are generalized to Vehicle.
4. Generalization (inheritance): CardPayment, CashPayment and OnlinePayment are generalized to Payment.
5. Composition: A Customer is composed of exactly one Licence, which has no existence apart from that Customer.
6. Composition: A Booking is composed of exactly one Invoice, which is discarded if the Booking is discarded.
7. Composition: A Booking is composed of zero to two ConditionReports, one at handover and one at return.
8. Aggregation: A Branch holds one or more Vehicles, each of which may be reassigned to another Branch.
9. Aggregation: A Branch employs one or more BranchStaff, who exist independently of the Branch.
10. Association: A Customer may place zero or more Bookings.
11. Association: A Booking must reserve exactly one Vehicle.
12. Association: A Booking is collected from exactly one Branch.
13. Association: A Booking is settled by one or more Payments.
14. Association: A BranchStaff may record zero or more ConditionReports.
15. Association: A Vehicle may undergo zero or more MaintenanceJobs.
16. Association: A MaintenanceTechnician may perform zero or more MaintenanceJobs.
17. Association: An Administrator defines zero or more Tariffs.
18. Dependency: A Booking verifies a Licence in order to establish eligibility, but does not own it.
19. Dependency: A Vehicle is priced by one or more Tariffs, each with its own effective date; the Tariff is used for calculation only.



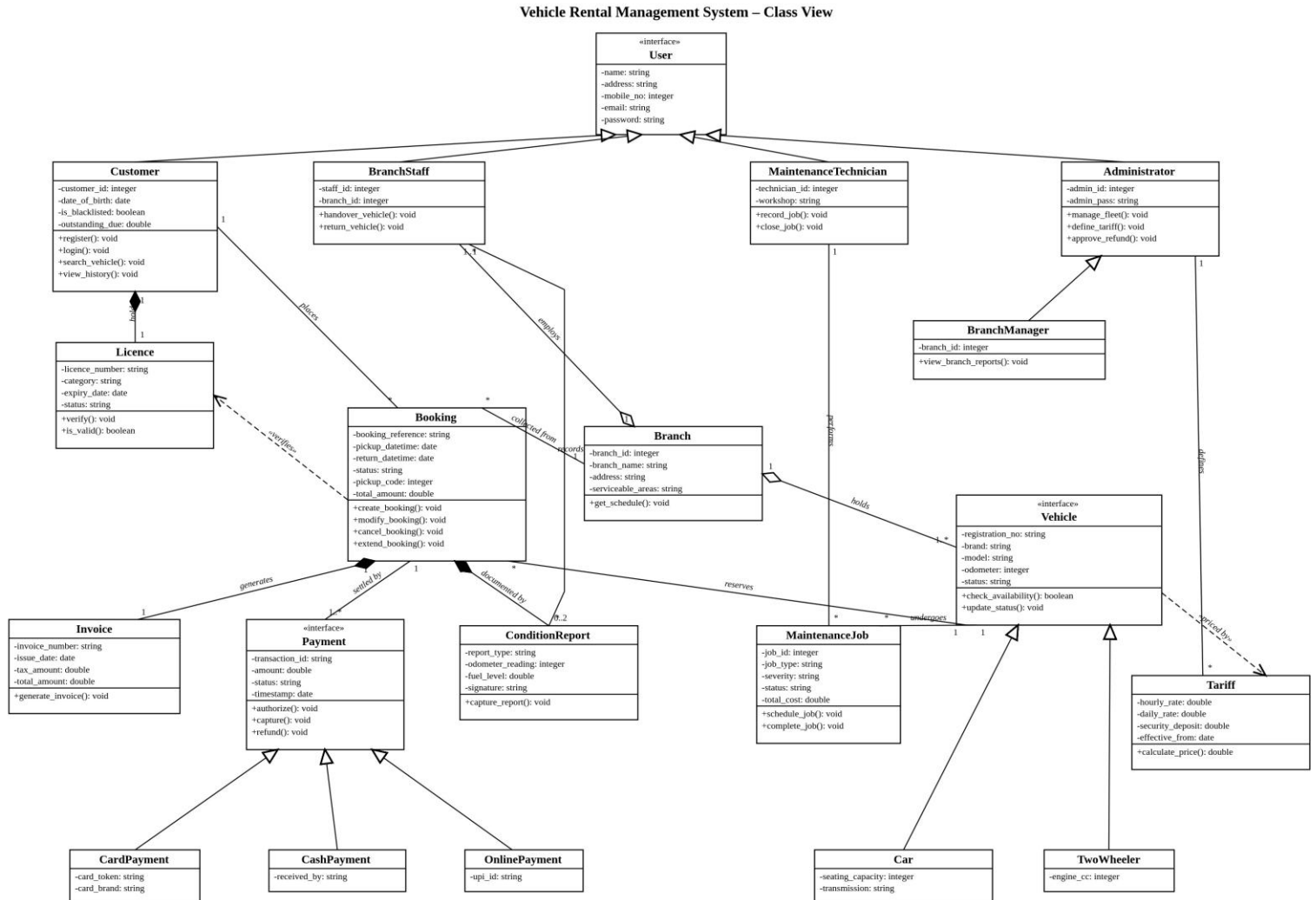
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[5] Class Diagram.





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Following are the code snippets that were written corresponding to each class and interface demonstrated in the above diagram -

User:

```
public interface User{
    String name;
    String address;
    int mobile_no;
    String email;
    String password;
}
```

Customer:

```
public class Customer implements User{
    private int customer_id;
    private Date date_of_birth;
    private boolean is_blacklisted;
    private double outstanding_due;

    public void register(){}
    public void login(){}
    public void search_vehicle(){}
    public void view_history(){}
}
```

Branch Staff:

```
public class BranchStaff implements User{
    private int staff_id;
    private int branch_id;

    public void handover_vehicle(){}
    public void return_vehicle(){}
}
```

Maintenance Technician:

```
public class MaintenanceTechnician implements User{
    private int technician_id;
    private String workshop;

    public void record_job(){}
    public void close_job(){}
}
```

Administrator:

```
public class Administrator implements User{
    private int admin_id;
    private String admin_pass;
```



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```
public void manage_fleet() {}  
public void define_tariff() {}  
public void approve_refund() {}  
}
```

Branch Manager:

```
public class BranchManager extends Administrator {  
    private int branch_id;  
  
    public void view_branch_reports() {}  
}
```

Licence:

```
public class Licence {  
    private String licence_number;  
    private String category;  
    private Date expiry_date;  
    private String status;  
  
    public void verify() {}  
    public boolean is_valid() {}  
}
```

Booking:

```
public class Booking {  
    private String booking_reference;  
    private Date pickup_datetime;  
    private Date return_datetime;  
    private String status;  
    private int pickup_code;  
    private double total_amount;  
  
    public void create_booking() {}  
    public void modify_booking() {}  
    public void cancel_booking() {}  
    public void extend_booking() {}  
}
```

Vehicle:

```
public interface Vehicle {  
    String registration_no;  
    String brand;  
    String model;  
    int odometer;  
    String status;  
  
    public boolean check_availability();  
    public void update_status();  
}
```

Car:



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```
public class Car implements Vehicle{  
    private int seating_capacity;  
    private String transmission;
```



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```
public boolean check_availability() {}  
public void update_status() {}  
}
```

Two-Wheeler:

```
public class TwoWheeler implements Vehicle {  
    private int engine_cc;  
  
    public boolean check_availability() {}  
    public void update_status() {}  
}
```

Branch:

```
public class Branch {  
    private int branch_id;  
    private String branch_name;  
    private String address;  
    private String serviceable_areas;  
  
    public void get_schedule() {}  
}
```

Tariff:

```
public class Tariff {  
    private double hourly_rate;  
    private double daily_rate;  
    private double security_deposit;  
    private Date effective_from;  
  
    public double calculate_price() {}  
}
```

Payment:

```
public interface Payment {  
    String transaction_id;  
    double amount;  
    String status;  
    Date timestamp;  
  
    public void authorize();  
    public void capture();  
    public void refund();  
}
```

Card Payment:

```
public class CardPayment implements Payment {  
    private String card_token;  
    private String card_brand;
```



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```
public void authorize() {}  
public void capture() {}  
public void refund() {}  
}
```

Cash Payment:

```
public class CashPayment implements Payment {  
    private String received_by;  
  
    public void authorize() {}  
    public void capture() {}  
    public void refund() {}  
}
```

Online Payment:

```
public class OnlinePayment implements Payment {  
    private String upi_id;  
  
    public void authorize() {}  
    public void capture() {}  
    public void refund() {}  
}
```

Invoice:

```
public class Invoice {  
    private String invoice_number;  
    private Date issue_date;  
    private double tax_amount;  
    private double total_amount;  
  
    public void generate_invoice() {}  
}
```

Condition Report:

```
public class ConditionReport {  
    private String report_type;  
    private int odometer_reading;  
    private double fuel_level;  
    private String signature;  
  
    public void capture_report() {}  
}
```

Maintenance Job:

```
public class MaintenanceJob {  
    private int job_id;  
    private String job_type;  
}
```



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```
private String severity;  
private String status;  
private double total_cost;  
  
public void schedule_job(){}  
public void complete_job(){}  
}
```

Conclusion:

By performing this experiment, we were able to understand various class diagram concepts used in object-oriented software design. We learned how to identify classes, attributes, methods, and the relationships between classes, including generalization (inheritance), association, aggregation, composition and dependency, with their respective multiplicities. We also successfully constructed a class diagram for our proposed Vehicle Rental Management System by identifying classes from the nouns in the problem description, defining their interactions, and representing them appropriately. Additionally, we translated the designed classes into corresponding Java class declarations, strengthening our understanding of how UML class diagrams can be used to plan and implement a software system effectively.